STROBE-MR Checklist

Study: Genetically higher BMD raises knee osteoarthritis risk yet protects against fracture on pleiotropy-robust evidence
Reporting guideline: STROBE-MR statement (Skrivankova VW et al. JAMA 2021;326(16):1614-1621, doi:10.1001/jama.2021.18236) - 20 main items, 30 subitems.

#	STROBE-MR item	Status	Location in manuscript
1	TITLE and ABSTRACT: Indicate Mendelian randomisation (MR) as the study's design in the title and/or the abstract if that is a main purpose of the study	Reported	Title ('...genetic correlation and bidirectional Mendelian randomisation across 13 skeletal phenotypes'); Abstract 'Design' paragraph
2	INTRODUCTION - Background: scientific background and rationale; what is the exposure; is a potential causal relationship plausible; justify why MR is a helpful method	Reported	Section 1, paragraphs 1-2: OA-BMD debate since Foss and Byers 1972, confounding and reverse causation in observational designs, why MR is informative
3	INTRODUCTION - Objectives: state specific objectives, including pre-specified causal hypotheses; state that MR intends to estimate causal effects under assumptions	Reported	Section 1, paragraphs 4-5 (three open questions; pre-specified predictions P1-P6 and falsifying tests F1-F10 in Table S11); MR as an assumption-bound estimator stated in Section 2 MR opening sentence and Section 4 limitations
4	METHODS - Study design and data sources: (a) setting; (b) participants, eligibility, sample size; (c) measurement, quality control and selection of genetic variants; (d) methods of assessment and diagnostic criteria; (e) ethics approval and consent	Reported	Section 2 Data sources: 13 GWAS with accessions (Refs 14-17, 23, 26), N and case counts, Table 1 ancestry and heritability; variant QC, palindromic exclusion and clumping in Section 2 LDSC and MR; DXA, radiographic and self-report phenotype definitions in Section 2 Data sources and Section 3; original ethics approvals and no new approval required, and the summary-statistic access date (17 July 2026), in Data availability
5	METHODS - Assumptions: explicitly state the three core IV assumptions (relevance, independence and exclusion restriction) as well as assumptions for additional or sensitivity analyses	Reported	Section 2 MR, opening sentence states all three assumptions verbatim. Assumptions of the added analyses are stated alongside them: MRlap (Section 2 Sample overlap) and MVMR including its no-measurement-error assumption and conditional F (Section 2 MVMR)
6	METHODS - Statistical methods: main analysis: (a) scale, units, model; (b) how variants were handled and weighted; (c) MR estimator, covariates, and whether the same covariate set was used in both samples; (d) missing data; (e) multiple testing	Reported	Section 2 Effect-scale reconstruction (per-SD BMD exposure; log-odds reconstruction for binary outcomes via 1/sqrt[K(1-K)] = 3.389; no re-centring); Section 2 MR (inverse-variance weighting; random-effects multiplicative convention SE_random = SE_fixed x sqrt[max(1, Q/df)]); Section 2 MVMR (MVMR-IVW and MVMR-Egger; no individual-level covariates, so the covariate set is identical across samples); no missing data were imputed - complete-case harmonisation with attrition reported in Figure S9; multiple testing: Bonferroni alpha = 0.05/78 and Benjamini-Hochberg FDR across the 78-pair matrix, alpha = 0.005 in MR
7	METHODS - Assessment of assumptions: describe any methods or prior knowledge used to assess the assumptions or justify their validity	Reported	Section 2 MR: Steiger directionality, per-SNP and aggregate F, R2_exposure, MR-Egger intercept, weighted median with bootstrap CIs, MR-PRESSO, leave-one-out, conservative 1-Mb clumping sensitivity. Section 2 Sample overlap: Burgess bound (<=2.6% BMD to OA, <=12% BMD to fracture) and MRlap. Section 2 Colocalisation: shared-causal-variant evidence at PP.H4 >= 0.8. Section 4 limitations: residual overlap and ancestry mismatch
8	METHODS - Sensitivity analyses and additional analyses: describe any sensitivity analyses or additional analyses performed	Reported	Section 2 MR, Sample overlap and MVMR: internal positive control (BMD to fracture calibrated against GEFOS), reverse MR, MRlap sample-overlap correction, MVMR conditioning on height and separately on BMI, leave-one-out, MR-PRESSO outlier removal, colocalisation; results in Section 3
9	METHODS - Software and pre-registration: (a) name statistical software and packages including version and settings; (b) state whether the protocol and details were pre-registered, and when and where	Reported	Section 2 opening paragraph: Python 3.13 with numpy, pandas, scipy and matplotlib; R 4.6.1 with MRlap 0.0.3.3. Data availability: the study pre-registration (P1-P6 as derived predictions, F1-F10 as registered tests with decision rules fixed before execution) is deposited with the public archive (Zenodo concept DOI 10.5281/zenodo.23400860)

#	STROBE-MR item	Status	Location in manuscript
10	RESULTS - Descriptive data: (a) numbers of individuals at each stage and reasons for exclusion, consider a flow diagram; (b) summary statistics; (c) heterogeneity across contributing studies; (d) for two-sample MR, similarity of variant-exposure associations between samples and extent of sample overlap	Reported	Table 1 (SNP-based heritability, N, cases, LDSC intercept); Table S14 ancestry composition; Figure S9 instrument attrition flow; Section 3 paragraph 1; similarity of variant-exposure associations assessed by cross-trait rho_GE (Figure 9) and overlap bounded in Section 2 Sample overlap
11	RESULTS - Main results: (a) variant-exposure and variant-outcome associations on an interpretable scale; (b) MR estimates with uncertainty on an interpretable scale, e.g. odds ratio per SD; (c) if relevant, translate relative risk into absolute risk; (d) consider plots to visualise results	Reported (11c N/A)	Table 2 (IVW beta, SE, p and random-effects p; MR-Egger slope and intercept; weighted median; OR per SD with 95% CI), Table 3, Figures 3-6 (forest, scatter, leave-one-out) and Section 3. Sub-item 11c was not translated to absolute risk: the source knee-OA GWAS defines no incidence window over which absolute risk could be computed
12	RESULTS - Assessment of assumptions: (a) report the assessment of the validity of the assumptions; (b) report additional statistics such as heterogeneity across variants (I2, Q statistic or E-value)	Reported	Section 3 and Table 2: MR-Egger intercept p > 0.24 at all five sites; Cochran Q and I2 = 68-86% forward versus 78-83% in the positive control; Steiger z = 3.05-6.11 forward and non-significant in reverse; MR-PRESSO distortion significant for 9/10 pairs with 0-3 outliers each
13	RESULTS - Sensitivity analyses and additional analyses: (a) robustness to violations of the assumptions; (b) other sensitivity analyses; (c) assessment of direction of causality, e.g. bidirectional MR; (d) comparison with non-MR estimates; (e) additional plots	Reported	Section 3: random-effects IVW; reverse MR; MVMR on height (Tables S9, S10) and BMI (Table S16); MRlap correction (Figure 10); comparison with the non-MR GEFOS estimate (OR 1.55) and with Hartley and Funck-Brentano (Figure S12); leave-one-out (Figure S2); multi-estimator robustness grid (Figure S11)
14	DISCUSSION - Key results: summarise key results with reference to study objectives	Reported	Section 4, paragraph 1
15	DISCUSSION - Limitations: taking into account validity of the IV assumptions, other sources of potential bias and imprecision; discuss direction and magnitude of any potential bias	Reported	Section 4 Strengths and limitations: ancestry mismatch versus European-only instruments; five prespecified sites deferred on compute; areal versus volumetric BMD; dependence of the significant count on the random-effects convention; reverse and OA-to-fracture power 0.14-0.26; residual overlap bounded at <=2.6% with its bias direction stated explicitly
16	DISCUSSION - Interpretation: (a) meaning - cautious overall interpretation in context and against other studies; (b) mechanism - biological mechanisms that could drive the causal relationship and whether gene-environment equivalence is reasonable; (c) clinical relevance	Reported	Section 4: Systemic Osteogenic Set-point posed as a testable hypothesis, subchondral bone remodelling within the osteochondral unit, hip-versus-knee site contrast, explicit statement that these estimates do not support changing prescribing practice and that a joint cost requires trials powered for incident OA at the joint of interest; Conclusion
17	DISCUSSION - Generalizability: (a) to other populations; (b) across other exposure periods/timings; (c) across other levels of exposure	Reported (17b N/A)	Section 4 limitations and Table S14: generalisation restricted to European-ancestry populations (87.31%-European knee-OA GWAS); F8 sex-stratified MR positive in both sexes. Sub-item 17b is not applicable because the germline instrument indexes a lifelong exposure with no definable timing; 17c is addressed by the per-SD scaling and by the MVMR attenuation ranges
18	OTHER INFORMATION - Funding: sources of funding and role of funders, including for the original databases	Reported	Funding: None; Acknowledgements state no funding was received. Source GWAS consortia acknowledged
19	OTHER INFORMATION - Data and data sharing: provide the data or report where and how they can be accessed, and provide the statistical code needed to reproduce the results	Reported	Data availability: all inputs are publicly published summary statistics (Refs 14-17, 23, 26) with accessions listed; every script and result table is archived at Zenodo concept DOI 10.5281/zenodo.23400860
20	OTHER INFORMATION - Conflicts of interest: all authors declare all potential conflicts of interest	Reported	Declaration of competing interest (none declared)

Overall: all 20 STROBE-MR items are addressed in the main manuscript and accompanying supplementary files. Two sub-items are not applicable to this design (11c, absolute-risk translation; 17b, timing of exposure) and are reported as

such; no main item is Not Applicable.